

4. AI Tutors

Rice Business 2026

Kerry Back
Rice Business

Workshop Overview

AUG 10
3:00–4:00

Intro to Claude

AUG 11
3:00–4:00

Working with Claude

AUG 12
3:00–4:00

Teaching Tools

AUG 13
3:00–4:00

AI Tutors

AUG 14
3:00–4:00

Student Projects

AUG 17
3:00–4:00

AI for Research

AUG 19
3:00–4:00

Personal Assistants

AUG 19
4:00–5:00

Overflow/Extra

Today's Session

1

Gemini Notebook



2

**Build Your Own RAG
Bot**



3

Skills as Tutors

Gemini Notebook (previously NotebookLM)

Free knowledge repository. Upload as many as 50 sources into each notebook. Get a chatbot that answers from sources, narrated slides, podcast-style audio, quizzes, ...

- Chatbot example: [Options textbook](#)
- Narrated slides example: [Kapadia-Ostdiek-Weston paper](#)
- Podcast example: [Put-call parity from textbook](#)

Chatbot is great. Videos are good for a short change of pace. We can specify **only use this part of the source** for video or podcast. Cannot revise directly but can regenerate with new instructions.

RAG Chatbots

Retrieval Augmented Generation

1

Prepare

Split your documents into chunks, turn each into a numeric “embedding,” and store them in a database



2

Retrieve

For a question, find the chunks most similar to it and hand them to the model



3

Answer

The model responds grounded in those chunks — and can cite which document each came from

Gemini Notebook uses RAG

Build Your Own

Chunk, embed, match

- Claude can embed your sources into a database
- Finding chunks that best match a prompt is a mechanical exercise that can be scripted

Read and interpret

- Reading the chunks and responding is AI. Could create a custom chatbot that makes API calls (more later).
- Or could use Claude as the AI. Create a Claude skill and include the database

Why Build Your Own

- More than 50 sources (research database — more later)
- More control over answers (via system prompt — more later)

Skills as Tutors

Get AI to Do the Teaching?



Our mission: To explore strange new worlds, to seek out new life and new civilizations, to boldly go where no one has gone before

Ideas

- Presentation feedback
- Writing feedback
- Practice cases/exercises with feedback
- Role-playing
- ???

Example

- Tell Claude to add the plugin marketplace github.com/kerryback/tutors
- Then tell Claude to install `binomial-tutor@kerryback-tutors`
- Start a new conversation
- Enter `/binomial-tutor`

Breakout

- Would tutors add value in your teaching? What types of tutors?
- As a school or in groups, how should we coordinate development of tutors? What incentives should there be, if any?